SOFTWARE DEVELOPMENT PROJECT SPECIFICATION

Course: Software Development Project (SWP391) | **Group Size:** 5 Students | **Project Code: SWP391-HOS-02**| **Lecturer**: ManhNC5@fpt.edu.vn |

Project Title: PhucNguyen Resort & Tour Hub (Integrated Hospitality Management System)

Domain Application: Hospitality, Food & Beverage (F&B), Tourism, E-Commerce

1. Problem Statement

Managing a modern luxury resort involves coordinating multiple disjointed services: Room Accommodations, On-site Dining (Restaurants), and Local Tour/Activity Bookings. Traditionally, resorts use separate software for front-desk operations, Point-of-Sale (POS) for restaurants, and manual spreadsheets for tour bookings. This fragmentation leads to double-bookings, lost revenue, poor customer experience (guests having to pay separately for every service), and data security vulnerabilities.

PhucNguyen Resort & Tour Hub is proposed to solve this by providing an "All-in-One" integrated web platform. It allows Guests to book rooms, order food to their rooms, and schedule tours through a single portal, while providing Resort Staff with a centralized dashboard to manage operations and generate a consolidated final invoice upon checkout.

2. Third-Party Tool Interface

To ensure industry standards, the system must integrate the following external APIs (simulated or real):

1. **Payment Gateway API (Stripe / VNPay Sandbox):**
   - **Function:** To securely process online deposits for room bookings and handle final checkout payments.
2. **Email/SMS Service API (SendGrid / Twilio):**
   - **Function:** To automatically send booking confirmations, QR code tickets for tours, and digital invoices to guests.
3. **Weather / Map API (OpenWeather / Google Maps):**
   - **Function:** Display real-time weather forecasts and interactive maps for outdoor Tour/Activity packages.

3. External Entity Specification (Actors)

1. **Guest (Customer):** The end-user who creates an account, books rooms, reserves restaurant tables, books tours, and views their consolidated bill.
2. **Receptionist (Front Desk):** Manages check-in/check-out processes, verifies guest identities, and processes final payments.
3. **Service Staff (F&B / Tour Guide):** Updates the status of food orders (Kitchen -> Served) or manages tour attendee lists.
4. **System Manager (Admin):** Manages the master data (Rooms, Menu items, Tour catalogs, Staff accounts) and views financial/occupancy dashboards.

4. Use Cases / User Stories (Distributed into 5 Modules)

This project is architected into 5 independent but interconnected modules. Each student must take ownership of one module (Full-stack: DB -> DAO -> Controller -> UI).

Module 1: Auth, Profiles & Master Data (Assigned to Student 1)

- **UC01:** As a guest, I want to register, verify my email, and securely log in.
- **UC02:** As a Guest, I want to manage my profile and upload my ID/Passport information (Requires explicit consent per Decree 356/2025).
- **UC03:** As an Admin, I want to manage staff accounts and assign roles (Receptionist, F&B Staff, Manager).
- **UC04:** As an Admin, I want to manage Room Categories, Restaurant Menus, and Tour Catalogs (CRUD operations with image uploads).
- **UC05:** As a Guest, I want to execute my "Right to Deletion" (Article 5, Decree 356/2025) to request the system to erase my personal data after my stay.

Module 2: Room Reservation & Front Desk - CORE 1 (Assigned to Student 2)

- **UC06:** As a Guest, I want to search for available rooms by date range, guest count, and room type.
- **UC07:** As a Guest, I want to book a room and pay a deposit securely.
- **UC08:** As a Receptionist, I want to view a daily dashboard of expected arrivals and departures.
- **UC09:** As a Receptionist, I want to perform Check-In for a guest (allocating a specific physical room number) and Check-Out.
- **UC10:** As a Receptionist, I want to update physical room statuses (e.g., Available, Occupied, Needs Cleaning, Maintenance).

Module 3: Restaurant POS & Room Service - CORE 2 (Assigned to Student 3)

- **UC11:** As a Guest (checked-in), I want to browse the restaurant menu and place a food order to be delivered to my room.
- **UC12:** As a Guest, I want to reserve a dining table for a specific date, time, and party size.
- **UC13:** As F&B Staff, I want to view a real-time list of incoming food orders and update their status (Pending -> Cooking -> Delivered).
- **UC14:** As F&B Staff, I want to attach the cost of a restaurant meal directly to a checked-in Guest's room bill (Charge to Room).

Module 4: Tour Booking & Feedback System (Assigned to Student 4)

- **UC15:** As a Guest, I want to browse local tours/activities, view itineraries, and check weather forecasts.
- **UC16:** As a Guest, I want to book a tour for a specific date and specify the number of participants.
- **UC17:** As a Tour Guide, I want to view the manifest (attendee list) for my assigned tours today.

- **UC18:** As a Guest, after checking out or completing a tour, I want to submit a 5-star rating and written review.
- **UC19:** As an Admin, I want to moderate guest reviews (Hide/Display) on the public portal.

Module 5: Consolidated Billing & Dashboards (Assigned to Student 5 - The Integrator)

- **UC20:** As a Receptionist, during Check-Out, I want the system to automatically generate a **Consolidated Invoice** that sums up the Room Cost, unpaid Food Orders, and unpaid Tour Bookings.
- **UC21:** As a Receptionist, I want to process the final payment (Cash/Credit) and mark the Consolidated Invoice as "Paid".
- **UC22:** As a Manager, I want to view a visual dashboard showing the "Room Occupancy Rate" (%) for the current month.
- **UC23:** As a Manager, I want to view a Revenue Dashboard breaking down income by source (Rooms vs. F&B vs. Tours).
- **UC24:** As a Manager, I want to export the monthly financial report to an Excel/PDF file.

5. Business Rules (Strict Constraints)

1. **Strict Concurrency (Double-Booking Prevention):** The system MUST guarantee that a specific room or a specific restaurant table cannot be booked by two different guests for overlapping dates/times.
2. **Consolidated Billing Constraint:** A guest **CANNOT** be checked out (UC09) if they have unpaid food orders (UC14) or tour bookings attached to their room. All outstanding charges must be settled into the Final Invoice (UC20).
3. **Cancellation Policy:** Room deposits are 100% refundable if canceled > 48 hours before the check-in date. No refund is issued if canceled within 48 hours.

6. References & Domain Knowledge (Mandatory Research for Students)

To build a realistic and industry-standard Property Management System (PMS) and Point of Sale (POS) integrated platform, students must research the following business domain operations and legal frameworks. Your Database and Workflow designs must reflect these standards.

A. Legal Compliance & Data Protection (Tuân thủ Pháp lý & Bảo mật)

1. **Vietnam Decree 356/2025/ND-CP on Personal Data Protection (Effective Jan 1, 2026):**
   - **Focus Area:** Handling Guest IDs, Passports, and payment info.
   - **Application (Module 1):** Strict compliance with Article 4 (Sensitive Data), Article 5 (Right to Deletion), and Article 6 (Explicit Consent). Passports must be encrypted. Guests must have a UI button to request data anonymization after check-out.
2. **Law on Residence 2020 (Luật Cư trú 2020) & Circular 55/2021/TT-BCA:**
   - **Focus Area:** Regulations on declaring temporary residence (Khai báo tạm trú) for domestic and foreign guests at lodging establishments.
   - **Application (Module 2):** The Check-in process must enforce the collection of required fields (Full name, DOB, ID/Passport number, Nationality) so the Hotel Manager can export a daily report to submit to the local police (Công an phường/xã).
3. **Vietnam Law on Tourism 2017 (Luật Du lịch 2017):**
   - **Focus Area:** Chapter VI - Rights and obligations of tourism service providers.
   - **Application (Module 4):** Tour itineraries must display mandatory safety warnings, insurance inclusions, and clear refund policies.

B. Hotel & Front Office Operations (Nghiệp vụ Tiền sảnh & Lưu trú)

4. **The Guest Cycle (AHLEI - American Hotel & Lodging Educational Institute):**

* **Focus Area:** The 4 stages of a guest's stay: *Pre-arrival* → *Arrival (Check-in)* → *Occupancy* → *Departure (Check-out).*

* **Application (Module 2 & 5):** Students must map their Use Cases strictly to this cycle. For example, a room's state machine must logically flow: `Vacant/Clean` → `Occupied/Clean` → `Occupied/Dirty` → `Vacant/Dirty` → (Housekeeping) → `Vacant/Clean`.

5. **Vietnam National Standard for Hotels (TCVN 4391:2015):**

* **Focus Area:** Classification and requirements for 4-star and 5-star resorts.

* **Application (All Modules):** Guides the database seeding and master data. A 5-star resort system must have functionalities to support Room Service (24/7), Spa, multiple F&B outlets, and Concierge tours.

C. F&B and Restaurant Operations (Nghiệp vụ Nhà hàng & POS)

6. **KOT (Kitchen Order Ticket) Workflow:**

* **Focus Area:** The universal operational flow of ordering food in a restaurant.

* **Application (Module 3):** When a guest orders food, the database must instantly create a KOT record with status `Pending`. The Kitchen UI dashboard updates it to `Preparing`, and Service Staff updates it to `Served`. Without this state-tracking, the POS module is invalid.

7. **Charge-to-Room (Folio Integration) Concept:**

* **Focus Area:** The mechanism where a guest signs a bill at the restaurant, and the POS system pushes the debt to the PMS (Front Desk) system.

* **Application (Module 3 & 5):** The database must have a `Guest_Folio` table linking the `Food_Order_ID` to the `Room_Booking_ID`.

D. Hospitality Finance & Reporting (Nghiệp vụ Kế toán Khách sạn)

8. **USALI (Uniform System of Accounts for the Lodging Industry):**

* **Focus Area:** The global standard for hotel accounting and financial reporting.

* **Application (Module 5):** The system's revenue dashboard must categorize income correctly into "Operating Departments": *Rooms Revenue*, *Food & Beverage Revenue*, and *Minor Operated Departments* (Tours/Activities).

9. **The Night Audit Process:**

* **Focus Area:** The end-of-day accounting process that rolls over the business date and posts room charges to guest folios.

* **Application (Module 5):** Students should simulate a "Daily Closing" function that calculates total occupancy rates and finalizes unpaid bills for the day.

(End of Project Specification)